

Chien Nguyen

Portfolio Website | chiennguyen.uw.dev@gmail.com | LinkedIn: chien-nguyen03 | GitHub: chien-se

EDUCATION

University of Washington Bothell

Expected Graduation: Dec 2026

Bachelor of Science in Computer Science & Software Engineering

Bothell, WA

- **Relevant Coursework:** Parallel & Distributed Computing, Operating Systems, Databases, Computer Architecture, Data Structures & Algorithms, Machine Learning, Web Development

Everett Community College

Graduated: September 2023

Associate in Arts and Sciences, Direct Transfer Agreement

Everett, WA

- Computer Science Honors Program (4.0 GPA)
- Vice President, Asian Student Union

PROJECTS

Real-Time Stock Analytics Pipeline | Kubernetes, Docker, Kafka, PostgreSQL

- Architected an event-driven, microservices pipeline ingesting 1,238+ streaming events/sec, deploying services as isolated Kubernetes pods with horizontal autoscaling to maintain throughput under variable load.
- Reduced cascading failure risk across 5+ pipeline components by implementing bulkhead isolation and health-check probes, achieving fault-tolerant recovery without full-system restarts.
- Instrumented all pipeline services with Prometheus metrics and Grafana dashboards, enabling visibility into Kafka consumer lag, pod health, and data throughput.

Distributed Semantic Search Engine | Python, PySpark, Databricks (AWS EC2/S3)

- Engineered distributed ETL workflows across an AWS-hosted Spark cluster processing 20M+ records, using mapInPandas for distributed batch inference and Parquet/Delta for columnar storage.
- Benchmarked local FAISS vector search against distributed cosine similarity via PySpark on Databricks, analyzing performance and scalability trade-offs across different dataset sizes.

Streaming Platform Session Management API | Python, FastAPI, PostgreSQL

- Implemented 4-gate session enforcement validating account status, device registration, location whitelist, and concurrent stream limits in a single atomic transaction before granting any stream.
- Designed a 6-table PostgreSQL schema with 3 composite indexes targeting active session and device lookups, eliminating full table scans on high-frequency query paths.
- Built real-time suspicious activity detection flagging users whose active session count exceeds their subscription plan's max streams, validated against a seeded dataset of 254 devices and 1,237 sessions.

LLM-Powered Knowledge Database Pipeline | Python, Claude API, Custom Claude Code Skills

- Built a fully local agentic pipeline that ingests raw PDFs, extracts concepts via Claude API (Haiku/Sonnet), and organizes them into a queryable wiki across 10 topic modules and 100+ concept files.
- Reduced Claude API token cost by 90% by implementing git-diff-based change detection that surgically processes only modified files instead of full folder scans.
- Engineered context-aware prompt caching and multi-model routing (Opus for planning, Haiku/Sonnet for execution) across 6 custom Claude Code skills supporting Q&A, gap analysis, and automated summarization.

SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, HTML, CSS

Data & ML: Pandas, NumPy, PyTorch, scikit-learn, Apache Spark, Databricks, Kafka, Hugging Face

Systems & Tools: PostgreSQL, FastAPI, Docker, Kubernetes, Git, AWS, Ollama, Claude Code